

Emanuele Rossi

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Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously social graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,261 citations; h-index 15 (July 2026).

Education

Imperial College London Oct 2020 – Feb 2024

Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein

- Research focused on **graph machine learning methods** for large real-world systems

University of Cambridge Oct 2018 – Jun 2019

Master in Advanced Computer Science, Distinction (4.0 GPA)

Imperial College London Oct 2015 – Jun 2018

Bachelor in Computer Science, First Class Honors (4.0 GPA)

Experience

Postdoctoral Researcher Rome (remote)

GLADIA Research Group [🔗](#), Sapienza University of Rome – With Prof. E. Rodolà Oct 2025 – Present

- Developing machine learning methods for animal communication.

Machine Learning Researcher NYC (remote)

VantAI Jan 2024 – Aug 2025

- **Co-led development of Neo-1 [🔗](#)**, a state-of-the-art all-atom latent diffusion model for multi-modal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

Machine Learning Researcher London

Twitter Jun 2019 – Feb 2023

- Developed **graph ML models** for Twitter's social graph and deployed a recommendation model that generated **1.2M additional daily follows**

Machine Learning Researcher London

Fabula AI Mar 2019 – Jun 2019

- Developed graph ML models for fake news detection as part of Fabula AI's small research team before **Twitter acquired the company [🔗](#)**

Software Engineering Intern California

Google Jul 2017 – Oct 2017

Selected Publications

[Full list](#)

Animal Communication & Audio

- *F. Mustun et al.* OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review 2026*.
- *E. Rossi and E. Rodolà.* **Communicating Sound Through Natural Language [🔗](#)**. *arXiv 2026*.
- *D. Marincione et al.* **Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models [🔗](#)**. *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)*.

Graph Learning

- *E. Rossi et al.* **Temporal Graph Networks for Deep Learning on Dynamic Graphs [🔗](#)**. *ICML GRL Workshop 2020*. **1,700+ citations** and **1,000+ GitHub stars [🔗](#)**.

- *E. Rossi**, *F. Frasca** et al. [SIGN: Scalable Inception Graph Neural Networks](#) [↗](#). *ICML GRL Workshop 2020*. **600+** citations and used in [production at Airbnb](#) [↗](#).
- *B. Chamberlain*, *S. Shirobokov*, *E. Rossi* et al. [Graph Neural Networks for Link Prediction with Subgraph Sketching](#) [↗](#). *ICLR 2023, Oral (top 5%)*. **200+** citations.

Leadership

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- **Co-Founder, LeadTheFuture** ([Forbes article](#) [↗](#)) *Jun 2018 – Mar 2023*
 - Built a non-profit mentoring community with more than **200 mentors** and **500 selected mentees**

Supervision and Mentoring

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- **Davide Marincione** – PhD Student @ Sapienza University of Rome *01/05/2025 – Present*
[Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models](#)
 - **Simone Antonelli** – ML Research Intern @ Amboss Technologies *01/10/2024 – Present*
[Predicting Link Closure in the Lightning Network with Temporal Graph Networks](#)
 - **Marco Pegoraro** – ML Research Intern @ VantAI *01/10/2024 – 01/04/2025*
[Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules](#)
 - **Harrison Rush** – ML Engineer @ Amboss Technologies *01/08/2024 – Present*
[MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning](#)
 - **Thomas Castiglione** – ML Researcher @ VantAI *01/03/2024 – 01/04/2025*
[Neo 1: a unified model for all-atom structure prediction and generation of all biomolecules](#)
 - **Andy Huang** – PhD Student @ Mila *01/03/2024 – 01/04/2025*
[TGB: Temporal Graph Benchmark for Machine Learning on Temporal Graphs](#)
[UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs](#)
 - **Vincent Davis** – ML Engineer @ Amboss Technologies *01/11/2023 – 01/05/2024*
[Channel Balance Interpolation in the Lightning Network via Machine Learning](#)

Research Grants & Fellowships

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- **ISCRA-C Research Computing Grant** – CINECA *02/07/2026*
[DOLPH-ID: Dolphin Emitter Identification from Whistle Acoustics](#) **≈€20,000**
 Allocation of 8,500 GPU-hours on Leonardo's NVIDIA A100 64 GB accelerators. equivalent 9 months
 - **Research Fellowship** – Sapienza University of Rome *01/04/2026*
 For postdoctoral research on machine learning for non-human communication. **€36,000** 12 months
 - **Google Cloud Research Credits** – Google *2022, 2023*
 GCP credits for PhD research. Received in two consecutive years. **\$1,000/year**

Awards

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- **Olav Beckmann Project Prize** – Imperial College London *01/01/2017*
 For outstanding second-year project work.
 - **Engineering Dean's List** – Imperial College London *2016, 2017*
 Top 10% of cohort. Received in two consecutive years.

- **G-Research Prize (Gloucester Research Ltd Prize at the time of award)** – Imperial College London
Awarded annually to up to 10 non-final-year Computer Science students for academic excellence. 01/01/2016
- **President’s Undergraduate Scholarship** – Imperial College London
Imperial’s most prestigious undergraduate distinction, awarded to fewer than 5% of the incoming cohort for academic excellence and potential. Funding: £3,000. 22/05/2015

Invited Talks

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- *Graph Machine Learning for Dynamic Financial Networks: From Foundations to the Lightning Network*
[Swissquote Workshop on Graph Learning in Financial Networks, EPFL](#)
[\[Slides \(HTML\)\]](#) [\[Slides \(PDF\)\]](#) 10/07/2026
 - *Machine Learning on Temporal Graphs: From Foundations to Emerging Frontiers*
Linguae-ML Seminar, Ecole Normale Supérieure (Paris) [\[Slides \(HTML\)\]](#)
[\[Slides \(PDF\)\]](#) 19/05/2026
 - *Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models*
Earth Species Project Community Deep Dive [\[Slides \(PDF\)\]](#) [\[Video\]](#) 30/01/2026
 - *A Primer on Graph Neural Networks*
COLT Seminar, University of Pompeu Fabra, Barcelona [\[Slides \(PDF\)\]](#) 22/11/2023
 - *Edge Directionality Improves Learning on Heterophilic Graphs*
Temporal Graph Reading Group [\[Slides \(PDF\)\]](#) [\[Video\]](#) 14/09/2023
 - *Edge Directionality Improves Learning on Heterophilic Graphs*
Explainability and Applicability of GNNs Workshop, Kassel 07/09/2023
 - *Improving Machine Learning at Twitter using Graphs*
Privacy Week [\[Video\]](#) 28/09/2022
 - *Learning on Graphs with Missing Node Features*
TigerGraph Reading Group 09/09/2022
 - *Learning on Graphs with Missing Node Features*
Learning on Graphs and Geometry Reading Group [\[Slides \(PDF\)\]](#) [\[Video\]](#) 01/03/2022
 - *Learning on Graphs with Missing Node Features*
Machine Learning Research Group, University of Oxford 24/02/2022
 - *Learning on Graphs with Missing Node Features*
University of Cambridge AI Research Group Talks 23/11/2021
 - *Graph Neural Networks with Almost No Features*
Toronto Machine Learning Summit 17/11/2021
 - *Machine Learning on Dynamic Graphs: Temporal Graph Networks*
MLOps World 2021 16/06/2021
 - *Machine Learning on Dynamic Graphs: Temporal Graph Networks*
Intel ML Seminar [\[Slides \(PDF\)\]](#) 26/04/2021
 - *Machine Learning on Dynamic Graphs: Temporal Graph Networks*
GNNSys’21 Workshop 09/04/2021
 - *Dynamic Graphs and Temporal Graph Networks*
University of Liverpool Networks and Distributed Computing Series [\[Slides \(PDF\)\]](#) 22/01/2021
 - *TGN: Temporal Graph Networks*
AISC [\[Slides \(PDF\)\]](#) [\[Video\]](#) 21/07/2020

Full Publication List

Under Review

- F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, **E. Rossi**, Y. Lakretz, G. G. de Polavieja and G. Sumbre. 2026. OpenWhistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review*. (0).
- H. Rush, V. Davis, S. Antonelli, V. Singh, J. Shrader and **E. Rossi**. 2026. MPFlow: Learning Budgeted Max-Flow Optimization on the Lightning Network with Deep Graph Reinforcement Learning. *arXiv*. (0).
- **E. Rossi** and E. Rodolà. 2026. Communicating Sound Through Natural Language. *arXiv*. (0).
- S. Antonelli, V. Davis, H. Rush, A. Potdevin, J. Shrader, V. Singh and **E. Rossi**. 2026. Predicting Channel Closures in the Lightning Network with Machine Learning. *arXiv*. (0).
- D. Kovtun, M. Akdel, A. Goncarenco, G. Zhou, G. Holt, D. Baugher, D. Lin, Y. Adeshina, T. Castiglione, X. Wang, C. Marquet, M. McPartlon, T. Geffner, **E. Rossi**, G. Corso, H. Stärk, Z. Carpenter, E. Kucukbenli, M. Bronstein and L. Naef. 2024. PINDER: The Protein Interaction Dataset and Evaluation Resource. *bioRxiv*. (37).

Peer-reviewed journal articles

- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2026. Training-Free Graph-Based Imputation of Missing Modalities in Multimodal Recommendation. *IEEE Transactions on Knowledge and Data Engineering*. (2).
- Y. Aluma, Z. Baron, R. Barrett, C. Baumgartner, G. Beguš, S. Bhattacharya, M. M. Bronstein, S. Dahan, O. Davis, S. de Haas, J. Defoe, J. DelPreto, R. Dessì, R. Diamant, J. Gatesy, K. George, S. Gero, D. Gibbons, D. Gibbons, S. Gil, S. Goldwasser, D. F. Gruber, O. Harve, A. Hernandez, M. Ishay, N. Jadhav, L. KC, A. Kenny, A. Leitao, M. Lucas, A. Maalouf, P. Malkin, Y. Mevorach, S. Pagani, O. Paradise, G. Petri, S. Poetto, **E. Rossi**, D. Rus, M. Salino-Hugg, A. Santoro, P. Sharma, D. Tchernov, A. Torralba, P. Tønnesen, D. M. Vogt and R. J. Wood. 2026. Description of a collaborative sperm whale birth and shifts in coda vocal styles during key events. *Scientific Reports*. (1).
- V. Singh, M. Khanzadeh, V. Davis, H. Rush, **E. Rossi**, J. Shrader and P. Lio. 2025. Bayesian Binary Search. *Algorithms*. (6).

Peer-reviewed conference proceedings

- V. Davis, V. Singh and **E. Rossi**. 2025. Channel Balance Interpolation in the Lightning Network via Machine Learning. *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*. (4).
- J. Gastinger, S. Huang, M. Galkin, E. Loghmani, A. Parviz, F. Poursafaei, J. Danovitch, **E. Rossi**, I. Koutis, H. Stuckenschmidt, R. Rabbany and G. Rabusseau. 2024. TGB 2.0: A Benchmark for Learning on Temporal Knowledge Graphs and Heterogeneous Graphs. *Advances in Neural Information Processing Systems*. (47).
- D. Malitesta, **E. Rossi**, C. Pomo, T. Di Noia and F. D. Malliaros. 2024. Do We Really Need to Drop Items with Missing Modalities in Multimodal Recommendation?. *Proceedings of the 33rd ACM International Conference on Information*. (50).
- S. Huang, F. Poursafaei, R. Rabbany, G. Rabusseau and **E. Rossi**. 2024. UTG: Towards a Unified View of Snapshot and Event Based Models for Temporal Graphs. *Learning on Graphs Conference (LoG)*. (20).
- S. Huang, F. Poursafaei, J. Danovitch, M. Fey, W. Hu, **E. Rossi**, J. Leskovec, M. M. Bronstein, G. Rabusseau and R. Rabbany. 2023. Temporal Graph Benchmark for Machine Learning on Temporal Graphs. *Advances in Neural Information Processing Systems*. (273).
- **E. Rossi**, B. Charpentier, F. Di Giovanni, F. Frasca, S. Günnemann and M. M. Bronstein. 2023. Edge Directionality Improves Learning on Heterophilic Graphs. *Learning on Graphs Conference (LoG)*. (202).

- B. Chamberlain, S. Shirobokov, **E. Rossi**, F. Frasca, T. Markovich, N. Hammerla, M. M. Bronstein and M. Hansmire. 2022. Graph Neural Networks for Link Prediction with Subgraph Sketching. *International Conference on Learning Representations (ICLR)*. **Oral (top 5%)**. (230).
- M. Mutti, R. De Santi, **E. Rossi**, J. F. Calderon, M. M. Bronstein and M. Restelli. 2022. Provably Efficient Causal Model-Based Reinforcement Learning for Systematic Generalization. *Proceedings of the AAAI Conference on Artificial Intelligence*. (36).
- **E. Rossi**, F. Monti, Y. Leng, M. M. Bronstein and X. Dong. 2022. Learning to Infer Structures of Network Games. *Proceedings of the 39th International Conference on Machine Learning, ICML*. (13).
- **E. Rossi**, H. Kenlay, M. I. Gorinova, B. Chamberlain, X. Dong and M. M. Bronstein. 2022. On the Unreasonable Effectiveness of Feature propagation in Learning on Graphs with Missing Node Features. *Learning on Graphs Conference (LoG)*. (196).
- B. Chamberlain, J. Rowbottom, M. I. Gorinova, S. D. Webb, **E. Rossi** and M. M. Bronstein. 2021. GRAND: Graph Neural Diffusion. *Proceedings of the 38th International Conference on Machine Learning, ICML*. (544).
- B. Chamberlain, **E. Rossi**, D. Shiebler, S. Sedhain and M. Bronstein. 2020. Tuning Word2vec for Large Scale Recommendation Systems. *RecSys - 14th ACM Conference on Recommender Systems*. (38).

Peer-reviewed workshop proceedings

- D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà and **E. Rossi**. 2025. Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models. *NeurIPS Workshop on AI for Animal Communication*. **Spotlight (top 15%)**. (3).
- J. Durairaj, Y. Adeshina, Z. Cao, X. Zhang, V. Oleinikovas, T. Duignan, Z. McClure, X. Robin, G. Studer, D. Kovtun, **E. Rossi**, G. Zhou, S. Veccham, C. Isert, Y. Peng, P. Sundareson, M. Akdel, G. Corso, H. Stärk, G. Tauriello, Z. Carpenter, M. Bronstein, E. Kucukbenli, T. Schwede and L. Naef. 2024. PLINDER: The Protein-Ligand Interactions Dataset and Evaluation Resource. *ICML ML for Life and Material Science Workshop*. (95).
- **E. Rossi**, B. Chamberlain, F. Frasca, D. Eynard, F. Monti and M. Bronstein. 2020. Temporal Graph Networks for Deep Learning on Dynamic Graphs. *ICML Workshop on Graph Representation Learning*. (1747).
- **E. Rossi**, F. Frasca, B. Chamberlain, D. Eynard, M. Bronstein and F. Monti. 2020. SIGN: Scalable Inception Graph Neural Networks. *ICML Workshop on Graph Representation Learning*. (600).
- **E. Rossi**, F. Monti, M. Bronstein and P. Lio. 2019. ncRNA Classification with Graph Convolutional Networks. *KDD Workshop on Deep Learning on Graphs*. (40).